

MANASI BHANGALE

AI/ML Engineer · Generative AI · Backend Development

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Summary

Computer Engineering graduate (2026) specialising in Generative AI, LLMs, RAG pipelines, and Computer Vision. Hands-on experience building production AI systems — multi-LLM orchestration, RAG-based document intelligence, offline voice agents, and real-time vision systems. Skilled in Python, LangChain, FastAPI, OpenCV, YOLOv8, and Transformer-based models. Committed to responsible AI: output evaluation, hallucination reduction, and safety compliance.

Work Experience

SiliconMount Tech Services

Jun 2026 – Present

AI/ML Engineer (Full-time)

Remote

- Promoted to full-time AI/ML Engineer following successful internship; continuing to lead development of the Multi-LLM Orchestration Chatbot and AI systems across enterprise tools.

SiliconMount Tech Services

Dec 2025 – May 2026

Software Developer Intern — AI/ML

Remote

- Built an **Intelligent Multi-LLM Orchestration Chatbot** that parses natural language queries and dynamically routes them to the appropriate LLM (OpenAI GPT / Ollama-hosted local models) based on query intent, domain, and required capability — deployed across four enterprise tools: *Asset Management*, *Smart Proctoring*, *CRM*, and *myBOQ*; enforced project-level **security scoping** so each module strictly accesses only its designated knowledge boundary, preventing cross-tool data leakage.
- Engineered an **AI Smart Proctoring System** using Python, FastAPI, OpenCV, MediaPipe, and YOLOv8 for real-time candidate monitoring — facial analysis, gaze tracking, posture detection, and multi-scenario suspicious activity detection with an automated alert generation pipeline.
- Performed business analysis for warehouse and inventory management systems, translating client workflows into functional specifications covering product master, purchase orders, batch/lot tracking, expiry management, and repacking operations.

Projects

AcademIQ | AI Academic Document Intelligence · RAG · LLM

GitHub ↗

- Designed a full **RAG pipeline** — overlapping chunking (400-char / 50-char overlap), semantic embeddings via **all-MiniLM-L6-v2**, **FAISS** vector indexing, and local LLM inference (Mistral, Phi-3 Mini via Ollama) — across a 500+ page academic document test set supporting 5 input formats (PDF, DOCX, TXT, image, URL).
- Evaluated AI-generated responses for **factual accuracy and hallucination reduction** vs. standalone LLM baselines; applied responsible AI practices (on-device inference for privacy); exposed document Q&A, quiz generation, and research analysis via Streamlit.

Onyx | Offline AI Voice Assistant · Multimodal · Edge AI

GitHub ↗

- Built a fully **offline, modular AI pipeline** — custom-trained **openWakeWord** for wake detection, Faster-Whisper for STT, **Qwen3:14B & Qwen2.5-VL** (Ollama) for NL understanding and multimodal screen analysis, and Coqui XTTS-v2 TTS with GPU/CPU/Piper fallback.
- Implemented **tool calling** (web, GitHub, Wikipedia search, Windows automation) and persistent conversational memory via **ChromaDB** with session-aware context retention — zero cloud dependency, demonstrating **edge AI deployment** capability.

Technical Skills

Generative AI & LLMs: RAG, Prompt Engineering, Prompt Optimisation, Model Fine-tuning, LLM Orchestration, Multi-model Routing, LangChain, Hugging Face Transformers, Ollama, Sentence Transformers, FAISS, ChromaDB

Machine Learning & CV: Deep Learning, NLP, Feature Engineering, Scikit-learn, TensorFlow, Keras, PyTorch, XGBoost, OpenCV, YOLOv8, MediaPipe, Object Detection, Image Segmentation, OCR, Video Analytics, Dataset Augmentation

Backend & MLOps: Python, FastAPI, Flask, REST APIs, Microservices, Docker, SQLAlchemy, MySQL, MongoDB, Git, Postman, Streamlit

Cloud & Software: AWS/GCP Fundamentals, Edge Device Inference, Vercel, API Integration, Swagger, Testing & Debugging

Data & Viz: Pandas, NumPy, EDA, Data Cleaning, Statistical Analysis, Power BI, Tableau

Responsible AI: Output Evaluation, Hallucination Reduction, Safety & Compliance, Privacy-preserving AI, On-device Inference

Education

Dr. Babasaheb Ambedkar Technological University, Lonere

2022 – 2026

B.Tech — Computer Engineering CGPA: 8.13

Lonere, Maharashtra

Certifications & Achievements

Certifications

- Generative AI with LangChain & Hugging Face — Udemy
- Data Science, ML, DL & NLP — Udemy
- Artificial Intelligence — AcmeGrade (IIT Bombay)

Achievements

- **LeetCode** — Rating 1,749 · Top 10.45% · 250+ problems solved
- Open Source Contributor — GSSoC 2025
- SIH 2025 — Cleared Internal Evaluation
- Technical Coordinator, GDSC DBATU